



Sandeep Kumar Kumawat

Roll No.: M25BBE007

Master of Technology

Bioscience and Bioengineering

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M. Tech. researcher in Bioscience and Bioengineering with hands-on experience in single-cell RNA-seq (scRNA-seq) and bulk RNA-seq analysis, Gene x Environment (G x E) interaction studies, and multi-omics data integration. Skilled in building end-to-end bioinformatics pipelines in Python and R covering quality control, differential expression, pathway enrichment, and correlation analysis. Experienced in translating high-dimensional transcriptomic data into biologically interpretable insights for disease biology and biomarker discovery.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M. Tech. (Bioscience & Bioengineering)	Indian Institute of Technology, Jodhpur	7.38 (Current)	Present
B.Tech. (Biotechnology)	ASPEE SHAKILAM Biotechnology Institute, NAU	8.02	2025
Senior Secondary	Board of Secondary Education, Rajasthan	79.6%	2016
Secondary	Board of Secondary Education, Rajasthan	79.0%	2014

RESEARCH EXPERIENCE

- Indian Institute of Technology, Jodhpur** Present
M. Tech. Thesis: Single-Cell-Based Approaches for GxE Analysis Jodhpur, India
 - Investigate cell-type-specific transcriptional responses associated with gene-environment interactions using single-cell transcriptomic approaches.
 - Execute end-to-end scRNA-seq data analysis workflows, encompassing quality control, preprocessing, clustering, dimensionality reduction, cell-type annotation, and downstream analysis.
 - Develop an analytical framework for high-dimensional single-cell data to extract biologically interpretable patterns and characterize GxE-associated cellular responses.
- Indian Institute of Technology, Jodhpur** May 2026 - July 2026
Summer Thesis Research: Computational Analysis of Lung Fibrosis Jodhpur, India
 - Integrated and batch-corrected three public bulk RNA-seq datasets (82 samples spanning Control, Early, and Advanced IPF) using ComBat, then ran limma differential expression testing across disease-stage comparisons.
 - Derived an 11-gene common DEG signature and characterized it via STRING protein-protein interaction and g: Profiler pathway enrichment analysis, identifying a core extracellular-matrix remodeling module.
 - Conducted a literature review comparing three independent beta-cell subtyping schemes in Type 2 Diabetes, motivating the unified re-annotation strategy now being pursued in ongoing thesis work.
- MicroCare Laboratory, Pathology Laboratory** Mar. 2025 - May 2025
Laboratory Intern
 - Performed microbiological and pathological laboratory procedures on 500+ clinical samples, including culture/media preparation and handling of blood, urine, and sputum specimens.
 - Conducted pathological analysis of clinical samples using standard laboratory procedures, simple microscopy, and compound microscopy.
 - Ran antimicrobial activity and antibiotic susceptibility testing across 20 antibiotics to evaluate microbial resistance patterns.
- NAU, Surat** July 2024 - DEC 2024
B. Tech. Thesis: Antibacterial Activity of Plant Extracts Surat, India
 - Prepared plant extracts using methanol, acetone, ethyl acetate, and petroleum ether and quantified extractive yield for *Murraya koenigii* and *Trigonella foenum-graecum*.
 - Evaluated antibacterial activity by agar well-diffusion assays against *B. subtilis*, *S. aureus*, *E. coli*, *S. typhi*, and *Serratia* across multiple concentrations.

PROJECTS

- Integrated Omics Analysis of IPF**
Bioinformatics Project
 - Integrated 3 GEO gene-expression datasets and applied data normalization and ComBat batch correction to enable robust comparative analysis.
 - Analyzed differential expression, immune-cell fraction, gene-cell fraction correlation, and protein-protein interaction networks to identify biologically relevant molecular patterns.
 - Applied Seurat, CIBERSORTx, GSVA, STRING, and Cytoscape, along with statistical testing, for multi-omics data analysis, visualization, and biological interpretation.

TECHNICAL SKILLS

- **Programming Languages:** Python, R
- **Bioinformatics & Genomics:** scRNA-seq, Bulk RNA-seq, Seurat, GEO Datasets, Differential Gene Expression, Pathway Enrichment, CIBERSORTx, GSVA, STRING, Cytoscape
- **Statistics & Data Science:** Correlation Analysis, Wilcoxon Test, Kruskal-Wallis Test, Data Normalization, ComBat, pandas, NumPy, SciPy, Matplotlib, Machine Learning
- **Visualization & Tools:** ggplot2, Seurat Visualization, Cytoscape, Git, Linux/Ubuntu, Jupyter Notebook, RStudio

Key COURSEWORK

- Bioinformatics / Computational Biology, Genomics and Transcriptomics, Molecular & Cell Biology, Statistics / Data Analysis, Imaging Cell Biology

PUBLICATIONS

- **Book Chapter:** “Solvent Extraction-Based Screening of *Murraya koenigii* for Antimicrobial Efficacy against Bacterial Pathogens.” Advances in AYUSH, Vol. 17

RESEARCH INTERESTS

- Single-Cell Genomics | Computational Biology | Gene x Environment (GxE) Analysis | multi-omics | Transcriptomics | Disease Biology | Biomarker Discovery